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The Blind Spot Paradox

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

Alexandre Boyer · Méline Goullou · Salomé Fomville · Ulysse Petit-Tichanné

Research Track, Topic 38, Central@Spillic. Supervisor: Raphaël Meuto

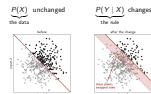
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The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

The Blind Spot Paradox

└ Concept drift: the rule changes, the data does not

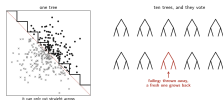
First, the problem. A model learns a rule from data. Then the world changes, and the rule becomes wrong: that is concept drift. In symbols, the inputs keep the same distribution, P of X , but the rule, P of Y given X , changes. Here is the simple case we simulate. Points are split by a line. At the change, the line moves. The points in the shaded band switch labels, so a model trained before the change now gets them wrong.



A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band.**

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└ Defence one: a model that repairs itself



This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

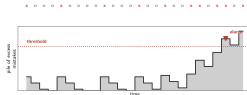
The first defence is a model that repairs itself: the Adaptive Random Forest. It is made of ten decision trees that vote. A single tree can only draw step-shaped borders, so it is never perfect. When one tree starts making too many mistakes, it is thrown away and a new one grows in its place.

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└ Defence two: a watchdog that raises the alarm

The second defence is a watchdog. Each mistake adds to a pile, each correct answer removes a little, never below zero. If the pile crosses a threshold, the alarm goes off and a human steps in. That alarm is all the human ever sees. So what happens when both defences run together?

the model answers, one observation at a time (x = wrong)



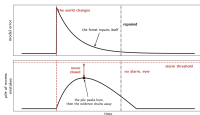
Two communities, two tools. What happens when both run at once?

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└ Put them together, and they cancel out

Here is what happens. The line moves, the mistakes start, and the pile begins to grow. But the forest replaces its trees, the mistakes stop, and the pile drains back to zero before it reaches the threshold. It is a race between the repair and the alarm. With a strict threshold, the repair almost always wins, and nobody is warned.

Put them together, and they cancel out



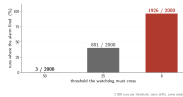
The repair has a date. The alarm never gets one. (schematic)

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└ Why not just lower the threshold?

So why not simply lower the threshold? We re-ran the paper's experiment at three thresholds, two thousand runs each. With the strictest one, the alarm went off three times in two thousand. With a lower one, it goes off almost every time. But a low threshold also goes off when nothing has changed, and an alarm that cries wolf ends up ignored. The paper concludes that no threshold is both trustworthy and sensitive.

Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

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└ Our contribution: when is the forest actually repaired?

To say who wins the race, you have to time the repair. The paper times it at the moment the first of the ten trees is replaced. But one tree out of ten is only the start of the repair. Wait until three quarters of the forest is renewed, and you wait fourteen times longer. The paper's stopwatch stops too early, and always in the forest's favour.



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└ Measuring the repair in mistakes, not in trees

So we also measure the repair in mistakes. Same change as before. The first tree goes after 86 observations. The error is back to normal only after 261, and three quarters of the forest after 1,225. The change costs about 38 extra mistakes. And the paper's stopwatch says nothing about that cost, at any of the eighteen change sizes we tested.

Measure	What it counts	Value
First tree replaced t_{first}	the date the paper uses	86
Error back to normal t_{norm}	the error settles back to its usual level	261
Forest 3/4 renewed t_{three}	8 trees out of 10 replaced	1,225
Excess mistakes $A(t)$	the extra errors the change caused	38
Watching's peak $t_{\text{peak}}(t)$	the highest point of the watching's pile	33

Same change as the previous slide, median of 100 runs. First three observations after the change, mistakes counted over the next 2,000 observations. Watching thresholds: 8, 25 and 50. Excess

The paper's date says nothing about the damage.

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└─Where we stand

Where we stand. Yes: with the paper's strict threshold, the alarm really stays silent, and we measured how often. But the paper's stopwatch stops at the first tree, so the forest looks faster than it is, and the paper's figures are too optimistic. A better stopwatch is still an open question: we are testing candidates, and none has passed yet. Thank you.

The blind spot is real

At the paper's strict threshold, the alarm fired **3 times in 2 000 runs**.

Stopwatch stops early

Up to **14x** too early: the repair looks faster than it really is.

Open question

What should replace that stopwatch? Candidates under test, none has passed yet.

Thank you for your attention.